

MATHEMATICS SUBJECTIVE TEST
Class X (Session 2026 – 27)

ANSWER KEY

SECTION-A

1. Since a, 7, b, 23 are in A.P.

$\therefore$ Common difference is same

$$\therefore 7 - a = b - 7 = 23 - b$$

Taking second and third terms, we get

$$b - 7 = 23 - b$$

$$\Rightarrow 2b = 30$$

$$\Rightarrow b = 15$$

Taking first and second terms, we get

$$\Rightarrow 7 - a = b - 7$$

$$\Rightarrow 7 - a = 15 - 7$$

$$\Rightarrow 7 - a = 8$$

$$\Rightarrow a = -1$$

Hence, $a = -1$, $b = 15$

2. From equation (2) : $x = 3 + \frac{y}{3}$ (i)

Substituting in equation (1) :

$$\frac{3 + \frac{y}{3}}{2} + \frac{2y}{3} = -1 \Rightarrow \frac{3}{2} + \frac{y}{6} + \frac{4y}{6} = -1$$

$$\text{From (i) : } x = 3 + \frac{-3}{3} = 3 - 1 = 2$$

$$\therefore x = 2, y = -3$$

3. Discriminant

$$(D) = (q - r)^2 - 4(p - q)(r - p) = (q + r - 2p)^2$$

Substituting $q + r = 2p$:

$$\Rightarrow D = (2p - 2p)^2 = 0$$

Therefore roots of the given equation are equal.

4. Given, A.P. is 65, 61, 57, 53,

Here, first term $a = 65$ and common difference, $d = -4$

Let the n^{th} term is negative.

Last term, $a_n = a + (n - 1)d = 65 + (n - 1)(-4) = 65 - 4n + 4 = 69 - 4n$,

which will be negative when $n = 18$

So, 18th term is the first negative term.

5. Here, $a = 15$ and $S_{15} = 750$

$$\because S_n = n/2[2a + (n - 1)d]$$

$$\therefore S_{15} = 15/2 [2 \times 15 + (15 - 1)d] = 750$$

$$\Rightarrow 15(15 + 7d) = 750$$

$$\Rightarrow 15 + 7d = 50$$

$$\Rightarrow 7d = 35$$

$$\Rightarrow d = 5$$

Now, 20th term $= a + (n - 1)d$

$$= 15 + (20 - 1)5$$

$$= 15 + 95 = 110$$

SECTION - B

6. **A.P. formation – $a_{15} - a_8 = 21$, $S_{10} = 55$**

$$a_{15} - a_8 = (a + 14d) - (a + 7d) = 7d = 21 \rightarrow d = 3$$

$$S_{10} = (10/2)[2a + 9d] = 55 \rightarrow 5[2a + 9d] = 55$$

$$2a + 27 = 11 \rightarrow 2a = -16 \rightarrow a = -8$$

A.P.: $-8, -5, -2, 1, 4, 7, \dots$

7. **Prove $2 + 3\sqrt{5}$ is irrational**

Assume $2 + 3\sqrt{5} = \frac{p}{q}$ (rational, $q \neq 0$)

$$\Rightarrow 3\sqrt{5} = \frac{p}{q} - 2 = (p - 2q)/q$$

$$\Rightarrow \sqrt{5} = (p - 2q)/3q \rightarrow \text{rational}$$

But $\sqrt{5}$ is irrational $\rightarrow$ Contradiction.

Hence $2 + 3\sqrt{5}$ is irrational

8. **Zeroes of $p(x) = x^2 - 15$**

$$x^2 - 15 = 0 \rightarrow x^2 = 15 \rightarrow x = \pm\sqrt{15}$$

Zeroes : $\alpha = \sqrt{15}$ and $\beta = -\sqrt{15}$

Verification :

$$\text{Sum of zeroes} = \sqrt{15} + (-\sqrt{15}) = 0 = -\frac{b}{a} = -\frac{0}{1} = 0$$

$$\text{Product of zeroes} = \sqrt{15} \times (-\sqrt{15}) = -15 = \frac{c}{a} = -\frac{15}{1} = -15$$

9. **$a_3 = 4$, $a_9 = -8$**

$$a + 2d = 4 \dots(i) \quad a + 8d = -8 \dots(ii)$$

(ii) – (i): $6d = -12 \rightarrow d = -2$

From (i) : $a = 4 - 2(-2) = 8$

(i) Which term is zero?

$a_n = 0$: $8 + (n-1)(-2) = 0 \rightarrow 10 - 2n = 0 \rightarrow n = 5$

5th term is zero

(ii) Find n when $S_n = -36$:

$(n/2)[2(8) + (n-1)(-2)] = -36$

$(n/2)[18 - 2n] = -36 \rightarrow n(9 - n) = -36$

$n^2 - 9n - 36 = 0 \rightarrow (n-12)(n+3) = 0$

$n = 12$ (rejecting $n = -3$)

10. Painting Competition Ages

$a = 8$ years, $d = 4$ months $= 1/3$ year, $S_n = 168$

$(n/2)[2 \times 8 + (n-1) \times (1/3)] = 168$

$(n/2)[(n-1)/3 + 16] = 168 \rightarrow n[(n+47)] = 1008$

$n^2 + 47n - 1008 = 0 \rightarrow (n - 16)(n + 63) = 0$

$n = 16$ (rejecting $n = -63$)

Eldest age $= a + (n-1)d = 8 + 15 \times (1/3) = 8 + 5$

Age of eldest participant = 13 years

SECTION - C

11. Right-angled Triangle – Area = 600 cm^2

Let altitude $= h$ cm, base $= (h + 10)$ cm

$(1/2) \times h \times (h+10) = 600 \rightarrow h(h+10) = 1200$

$h^2 + 10h - 1200 = 0$

$(h + 40)(h - 30) = 0 \rightarrow h = 30$ (rejecting $h = -40$)

Three dimensions :

Altitude $= 30$ cm

Base $= 30 + 10 = 40$ cm

Hypotenuse $= \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)} = \sqrt{2500} = 50 \text{ cm}$

Altitude = 30 cm, Base = 40 cm, Hypotenuse = 50 cm

12. Fraction Problem

Let fraction $= x/y$

Setting up equations :

$(x+3)/(y+3) = 5/6 \rightarrow 6x + 18 = 5y + 15 \rightarrow 6x - 5y = -3 \dots(i)$

$(x+2)/(y+2) = 9/11 \rightarrow 11x + 22 = 9y + 18 \rightarrow 11x - 9y = -4 \dots(ii)$

Solving :

Multiply (i) $\times 9$: $54x - 45y = -27 \dots(iii)$

Multiply (ii) $\times 5$: $55x - 45y = -20$... (iv)

(iv) - (iii) : $x = 7$

Into (i) : $42 - 5y = -3 \rightarrow y = 9$

Verification : $(7 + 3)/(9 + 3) = 10/12 = 5/6 \checkmark$ $(7+2)/(9+2) = 9/11 \checkmark$

Original fraction = $7/9$

13. Potato Race – Bucket at start, 1st potato 5m away, gap = 3m

Distance of kth potato from bucket = $5 + (k - 1) \times 3 = (3k + 2)$ m

Round trip for kth potato = $2 \times (3k + 2) = (6k + 4)$ m

(i) Distance for 1st potato :

$K = 1$: $6(1) + 4 = 10$ m

10 m

(ii) Distance for 2nd potato :

$K = 2$: $6(2) + 4 = 16$ m

16 m

(iii) (a) Total distance :

Distances : 10, 16, 22, 28, 34, 40, 46, 52, 58, 64 $\rightarrow$ AP with $a = 10$, $d = 6$, $n = 10$

$S_{10} = (10/2)[2 \times 10 + 9 \times 6] = 5 \times [20 + 54] = 5 \times 74$

Total distance = 370 m

(iii) (b) Time taken :

Time = Distance $\div$ Speed = $370 \div 5$

Time = 74 seconds

14. Kolam Dots — 4, 8, 12, ...

(i) Show it is an A.P. :

$a_1 = 4$, $a_2 = 8$, $a_3 = 12$

$a_2 - a_1 = 4$, $a_3 - a_2 = 4 \rightarrow$ constant difference $\rightarrow$ forms A.P.

First term $a = 4$, Common difference $d = 4$

(ii) n^{th} term :

$a_n = a + (n-1)d = 4 + (n-1) \times 4$

$a_n = 4n$

(iii) Find n when total dots = 220 :

$S_n = 220$

$(n/2)[2 \times 4 + (n-1) \times 4] = 220 \rightarrow (n/2)[4n+4] = 220$

$2n(n+1) = 220 \rightarrow n(n+1) = 110$

$n^2 + n - 110 = 0 \rightarrow (n+11)(n-10) = 0$

$n = 10$ squares (rejecting $n = -11$)

15. Multistorey Building Stairs – 10, 24, 38, ...

(i) Does 10, 24, 38, ... form an A.P.?

$$24 - 10 = 14 \quad 38 - 24 = 14 \rightarrow \text{common difference is constant}$$

Yes, it is an A.P. with $a = 10$, $d = 14$

(ii) Total stairs from ground to 11th floor:

$$S_{11} = (11/2)[2 \times 10 + 10 \times 14] = (11/2)[20 + 140]$$

$$= (11/2) \times 160 = 11 \times 80$$

$$\text{Total stairs} = 880$$

(iii) (a) Total stairs for water supply to 6 floors :

$$\text{Stairs from ground to floor } k : S_k = k(3 + 7k)$$

$$\text{Round trip to floor } k = 2 \times S_k :$$

$$\text{Floor 1: } 2 \times 10 = 20 \mid \text{Floor 2: } 2 \times 34 = 68 \mid \text{Floor 3: } 2 \times 72 = 144$$

$$\text{Floor 4: } 2 \times 124 = 248 \mid \text{Floor 5: } 2 \times 190 = 380 \mid \text{Floor 6: } 2 \times 270 = 540$$

$$\text{Total} = \Sigma(k=1 \text{ to } 6) 2S_k = 20 + 68 + 144 + 248 + 380 + 540$$

$$\text{Total stairs climbed} = 1400$$

OR

(iii) (b) If total = 380 stairs, find floor :

$$\text{The round trip to floor } n = 2 \times S_n = 380 \rightarrow S_n = 190$$

$$S_n = n(3 + 7n) = 190 \rightarrow 7n^2 + 3n - 190 = 0$$

$$n = \left(-3 \pm (\sqrt{9 + 5320}) \right) / 14 = \left(-3 \pm \sqrt{5329} \right) / 14 = (-3 \pm 73) / 14$$

$$n = 70/14 = 5$$

The person supplied water cans till the 5th floor.

